

Andrew Rodowicz

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SUMMARY

Data Scientist with hands-on experience in Python, SQL, machine learning, and marketing analytics. Proven ability to analyze complex datasets, optimize business performance, and translate insights into actionable strategies. Background spans digital marketing, sales leadership, and operations, bringing strong business context, communication skills, and analytical problem-solving to data-driven teams.

SKILLS

Python | SQL | Scikit-Learn | Data Analysis | Data Visualization | Jupyter | A/B Testing | Churn Analysis | EDA | Power BI | Matplotlib | Seaborn | Data Pipelines | Data Modeling

TECH PROJECTS

Oil Well Profit Optimization & Risk Analysis | [link](#)

Engineered an end-to-end predictive modeling and statistical risk analysis pipeline to maximize expected profit for oil well development under uncertainty and financial constraints.

- Trained and evaluated linear regression models in Python (pandas, scikit-learn) to predict oil reserve volumes across multiple regions, using a 75/25 train-validation split and RMSE for model assessment.
- Formulated a constrained optimization approach to rank model predictions and select top-performing candidates, maximizing expected profit under fixed budget and sample-size constraints.
- Applied non-parametric bootstrapping (1,000 resamples) to quantify uncertainty in profit estimates, computing confidence intervals and probabilistic loss metrics.
- Conducted risk-aware model comparison to identify the region with the highest expected return while satisfying statistical loss thresholds ($<2.5\%$).

Telecom Customer Churn Prediction | [link](#)

Developed an end-to-end machine learning pipeline to analyze customer churn behavior and predict customer attrition for a telecommunications provider using customer contract, internet service, phone service, and demographic datasets.

- Performed exploratory data analysis (EDA) to identify churn patterns across monthly charges, tenure, service usage, and customer contract characteristics
- Engineered predictive features including customer tenure metrics, service count indicators, monthly charge ratios, and churn risk segmentation variables
- Built and evaluated multiple classification models including Logistic Regression, Random Forest, CatBoost, LightGBM, and Gradient Boosting
- Applied RandomizedSearchCV with 5-fold cross-validation for hyperparameter optimization and model selection
- Selected and validated a final optimized model to improve customer churn prediction accuracy and support retention-focused business decision-making
- Created visualizations to communicate customer behavior trends and model performance insights

IMDB Review Sentiment Analysis | [link](#)

Developed a natural language processing (NLP) machine learning model to classify movie reviews as positive or negative by preprocessing text data, extracting linguistic features, and evaluating multiple classification algorithms.

- Processed and analyzed large-scale IMDB movie review text data to build a sentiment classification model for binary text prediction
- Performed extensive text preprocessing, including tokenization, normalization, stopwords removal, and feature preparation for machine learning workflows
- Conducted exploratory text analysis to identify sentiment distribution patterns and review characteristics
- Applied feature extraction techniques including TF-IDF vectorization to convert unstructured text into machine learning-ready numerical representations
- Built and compared classification models including Logistic Regression and other supervised learning algorithms for sentiment prediction
- Optimized classification performance through preprocessing refinement and model parameter tuning
- Visualized sentiment classification results and model evaluation metrics to communicate predictive performance
- Developed a reproducible NLP workflow for text-based sentiment analysis applications

EXPERIENCE

Proclaim Promotions | Digital Marketing Analyst | 04/25 - Present | Concord

Skills Used: AB Testing, Data Analysis, Business Analytics

- Analyze marketing and customer data to optimize digital campaigns for automotive dealerships, improving engagement and revenue.
- Evaluate KPIs, customer behavior, and A/B test results to refine multi-channel marketing strategies.
- Develop data-informed targeting and segmentation frameworks to improve conversion performance.
- Partner with sales and operations teams to align insights with business objectives and growth initiatives.

American Income Life | Supervising Agent | 01/21 - 04/25 | Remote

Skills Used: KPI Design, Data Driven Decision Making, Statistical Thinking

- Led and developed teams using performance metrics and sales KPIs to improve productivity and outcomes.
- Applied quantitative and qualitative analysis to client needs assessments and policy recommendations.
- Leveraged CRM data to improve client retention, engagement, and strategic outreach
- Monitored trends to refine sales strategies and consistently meet or exceed targets.

EDUCATION

TripleTen | Data Science

- Project-based program covering Python, SQL, machine learning, Tableau, A/B testing, and predictive modeling. including revenue analysis, customer churn prediction, and regional performance comparison.